IGSW: Multi-Weather Urban Scene Reconstruction with Intrinsic-Decoupled Gaussian Splatting

***Abstract*—Weather editing of dynamic driving scenes must change appearance while preserving scene structure and temporal continuity. Image editors provide realistic weather cues, but fitting their outputs can transfer structural errors into reconstructed scenes. We present IGSW, a two-stage dynamic Gaussian framework for rain and snow editing. First, shared albedo and normals encode stable background content, while weather-specific irradiance and residuals model appearance changes. Pooled intrinsic and edge features provide soft supervision from noisy inverse-rendering estimates. Second, a structure-preservation gate combines semantic and intrinsic conflicts to protect unreliable teacher regions, while per-Gaussian transport refines weather appearance with geometry and shared content fixed. The resulting scene renders directly without image teachers at inference. Experiments on nuScenes and Waymo show improved content preservation, semantic consistency, and temporal stability, alongside competitive weather-direction alignment. Ablations identify complementary roles for intrinsic modeling and protected refinement in balancing weather strength with scene fidelity.**

***Index Terms*—Gaussian splatting, intrinsic decomposition, weather editing, dynamic scene reconstruction, driving simulation.**

I. Introduction

WEATHER editing can extend captured driving scenes to conditions that are difficult to record with matched viewpoints and traffic. Rain changes illumination and road reflections; snow changes surface appearance and contrast. Yet roads, buildings, and traffic participants must retain their spatial layout and motion. A useful simulation therefore requires more than convincing individual images: weather appearance must remain coherent when the same dynamic scene is rendered across cameras and time.

3D Gaussian Splatting [1] and dynamic urban representations [3], [4] support efficient reconstruction and rendering, but observations of one condition do not determine how the scene should look in another. Text-guided editors such as Qwen-Image [18] supply target-weather images for scene optimization. WeatherEdit [6] and WeatherCity [5] combine such appearance priors with scene-level modeling, moving weather synthesis beyond independent two-dimensional edits. Their success motivates a closer examination of what a scene should learn from its image teachers.

Desired weather changes and unwanted content changes often coexist in a teacher image. Dark clouds may strengthen rainy atmosphere while also covering a building that should remain visible. Reflections may become more convincing while vehicle details drift between frames. Fitting these changes together can transfer teacher errors into Gaussian appearance and weaken structural detail. Conversely, strongly anchoring the result to original RGB values can preserve content at the cost of weather expression. This tension raises two connected questions: *which scene factors should vary with weather, and which teacher regions provide reliable supervision?*

Intrinsic decomposition offers a way to distinguish these roles. Albedo and normals provide cues to material and structure, whereas irradiance describes lighting variation. IntrinsicWeather [19] estimates such factors through generative inverse rendering. These estimates, however, are image-space predictions rather than exact physical measurements. Direct pixel matching can reproduce their local errors, and decomposition alone does not establish a representation shared by different views. We therefore treat intrinsic maps as soft priors for a renderable scene, with original observations supplying the reference for stable content.

We introduce IGSW (IntrinsicGSWeather), a two-stage framework built around this separation. Stage I decodes background Gaussian appearance into shared albedo and normals, together with weather-specific irradiance and residuals. A parameter-free encoder extracts pooled values and edges from rendered and estimated intrinsic maps. Shared-content features are aligned with the original intrinsic reference, while irradiance features follow the original-to-weather illumination change. This gives content preservation and weather adaptation distinct learning signals without requiring exact agreement with noisy intrinsic pixels.

Stage II strengthens weather appearance through structure-preserving refinement. Semantic conflicts and albedo/normal discrepancies identify teacher regions that should not be copied directly. Reliable regions retain full teacher supervision; conflicting regions preserve original spatial content while receiving teacher color statistics. After a short original-view recovery, geometry and shared content are fixed. Weather-specific heads and point-attached transport variables then adapt local illumination and residual appearance. Supervision reliability and appearance flexibility are thus addressed together, rather than asking a single RGB objective to resolve their conflict.

The optimized output remains a dynamic Gaussian scene rendered from camera, time, and weather conditions, without an image editor or teacher-image compositing at inference. Experiments on nuScenes and Waymo assess rain and snow editing using content, weather-direction, semantic, and temporal measures. Quantitative comparisons and ablations show how intrinsic modeling and protected refinement balance weather expression with scene fidelity.

Our contributions are threefold:

- An intrinsic-decoupled Gaussian representation with soft feature priors that assigns shared content and weather-dependent appearance distinct representational and supervisory roles.



Fig. 1. Overview of IGSW framework for multi-weather urban scene reconstruction.

- Structure-preserving weather refinement that combines reliability-aware teacher supervision with per-Gaussian transport while keeping geometry and shared content fixed.
- Full-sequence rain and snow evaluations on nuScenes and Waymo, with module and component ablations examining content preservation, weather expression, and temporal stability.

II. Related Work

A. Image and Video Weather Editing

Image translation and restoration methods [11], [12] model weather-dependent changes in two-dimensional observations. Text-guided approaches, including InstructPix2Pix [13], ControlNet [14], Prompt-to-Prompt [15], and Qwen-Image [18], offer more flexible control over target appearance. TokenFlow [16] and FRESCO [17] propagate correspondences to improve video consistency. However, temporal correspondences within an edited sequence do not by themselves establish a common scene across cameras. Moreover, a visually plausible edit may alter a building or vehicle that should remain unchanged. IGSW uses image editing for weather supervision, but learns the output in a shared dynamic scene and separates reliable appearance changes from teacher-induced content errors.

B. Weather-Aware Three-Dimensional Scenes

NeRF [2] and 3D Gaussian Splatting [1] provide implicit and explicit scene representations; Street Gaussians [3] and DrivingGaussian [4] extend reconstruction to dynamic driving environments. Weather synthesis adds appearance changes beyond captured observations. ClimateNeRF [10], WeatherEdit [6], Weather-Magician [7], and WeatherCity [5] explore scene-level weather modeling, while RainyGS [8] and Let It Snow! [9] target specific rain and snow phenomena. A common scene helps coordinate rendering across views, but does not ensure that edited-image supervision is structurally correct. In particular, RGB fitting couples lighting changes with local teacher errors. IGSW builds on dynamic Gaussian scene organization and introduces distinct intrinsic appearance roles, followed by protected supervision for weather refinement. Its focus is background appearance transfer rather than an additional particle-rendering pass.

TABLE I
Comparison of method categories and representation capabilities.

Method	Type	Intrinsic decomp.	3D/4D scene	Temporal consistency
Qwen-Image	Frame editing	×	×	×
FRESCO	Video editing	×	×	✓
IntrinsicWeather	Intrinsic image editing	✓	×	×
WeatherEdit	3D/4D weather	×	✓	✓
WeatherCity	4D weather	×	✓	✓
IGSW (Ours)	Intrinsic 4DGS	✓	✓	✓

C. Intrinsic Decomposition and Generative Inverse Rendering

Intrinsic Image Diffusion [20], IntrinsicEdit [21], RGB↔X [22], Uni-Renderer [23], and DiffusionRenderer [24] use generative priors to infer or manipulate material and illumination factors. IntrinsicWeather [19] applies intrinsic-space reasoning to weather editing. These factors provide a more structured reference than RGB alone, but single-image inverse rendering remains ambiguous, and independently estimated maps may disagree across weather or views. Treating every intrinsic pixel as ground truth can therefore introduce another source of supervision error. IGSW instead compares pooled intrinsic and edge features, anchors stable content to the original observations, and uses irradiance changes to guide weather appearance. The intrinsic estimates also help detect unreliable teacher regions; they are not required at inference.

III. Method

Given calibrated multi-view observations $\{I^o_{t,v}\}$, IGSW learns a dynamic scene that renders the original condition and a target weather. Here t indexes time, v indexes the camera, and o denotes the original condition. Figure 2 summarizes two stages. Intrinsic Gaussian modeling first assigns stable content and weather variation distinct representations. Structure-preserving refinement then strengthens teacher appearance

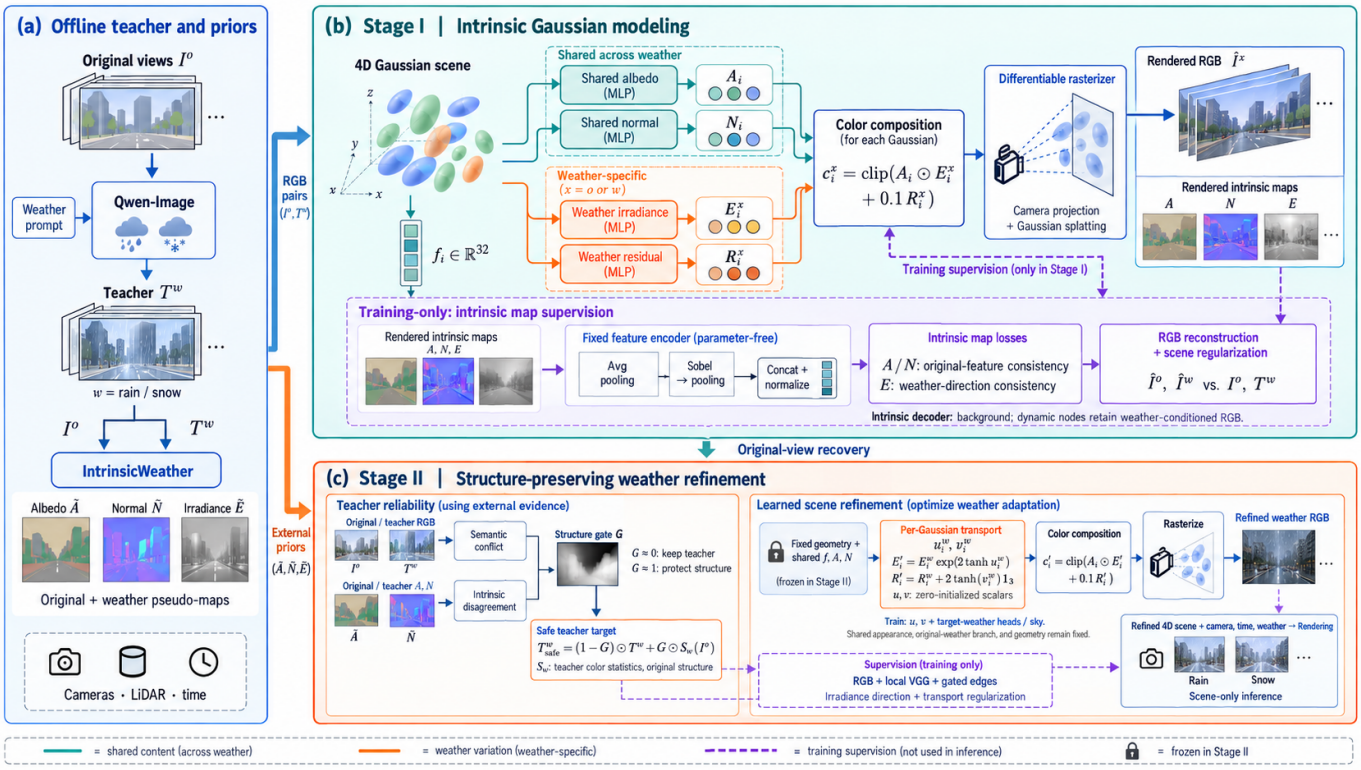


Fig. 2. IGSW overview. Original and edited images provide RGB supervision and intrinsic priors. Stage I learns shared albedo/normals and weather-specific irradiance/residuals. Stage II combines a structure-preservation gate with Gaussian weather transport to refine appearance without changing shared content.

transfer without allowing unreliable teacher regions to overwrite shared content. Scene organization, dynamic objects, sky, and camera correction follow WeatherCity [5].

A. Text-Guided Weather Teachers

Qwen-Image [18] produces a teacher $T^w_{t,v} = \mathcal{E}(I^o_{t,v}, p_w)$, where $\mathcal{E}$ is the image editor, $w \in \{\mathrm{rain}, \mathrm{snow}\}$ is the target weather, and p_w is its prompt. Prompts request weather changes while preserving layout, objects, and viewpoint (Supplementary Table S1). The teachers retain the original time/camera indexing and provide the desired illumination and atmosphere. Their correspondence is index-based, however: generated clouds or surface details can still displace scene content. We use these images as appearance guidance rather than assuming that every edited region is structurally reliable. Below, we omit t, v when the observation is unambiguous.

B. Intrinsic Priors from Inverse Rendering

The diffusion-based inverse renderer in IntrinsicWeather [19], built on Stable Diffusion 3.5, decomposes both I^o and T^w into albedo $\tilde{A}^x$, normals $\tilde{N}^x$, and irradiance $\tilde{E}^x$, where $x \in \{o, w\}$. Tildes denote offline pseudo-maps. Albedo emphasizes base surface color, normals provide structural cues, and irradiance captures lighting and shading. These factors supply complementary references for changes entangled in RGB.

Original albedo and normals anchor content that should remain stable across weather; original irradiance defines the starting point for an illumination change toward the teacher. Both conditions are decomposed, but edited albedo and normals are not imposed as new content targets. They also provide evidence for detecting teacher conflicts in Stage II. Because the estimates can contain local errors, supervision compares their pooled features rather than enforcing pixel-wise equality.

C. Intrinsic-Decoupled Gaussian Appearance

Each background Gaussian has center $\mu_i \in \mathbb{R}^3$, scale $s_i \in \mathbb{R}^3$, rotation quaternion $q_i \in \mathbb{R}^4$, opacity $\alpha_i \in [0, 1]$, and a learned appearance feature $f_i \in \mathbb{R}^{32}$. The feature is a latent code, not a concatenation of predefined material values. Gradients from rendered RGB and intrinsic features optimize it jointly with the scene. Shared and weather-specific multilayer perceptrons (MLPs) then decode this common code into attributes with different roles.

1) Shared Albedo and Normals: Shared heads predict $A_i = \sigma(h_A(f_i))$ and $N_i = \tanh(h_N(f_i))$, where h_A, h_N are MLPs and σ is the sigmoid. Both outputs have three channels and are shared across weather. Using the same heads and feature prevents an unrelated base color or normal attribute from being assigned to each weather. Albedo anchors base color; normals provide auxiliary structural cues, supervised in camera coordinates but not directly used in color composition. They are learned attributes rather than normals analytically derived from Gaussian covariance.

2) Weather-Specific Irradiance and Residuals: Separate heads predict $E_i^x = \sigma(h_E^x(f_i))$ and $R_i^x = \tanh(h_R^x(f_i))$. The irradiance branch captures illumination changes, and the

residual accommodates appearance not explained by their multiplicative interaction with albedo. Gaussian color is

$$c_i^x = \mathrm{clip}(A_i \odot E_i^x + \lambda_R R_i^x, 0, 1)\,, \tag{1}$$

where $\odot$ is channel-wise multiplication, $\lambda_R = 0.1$ controls residual amplitude, and clip bounds RGB values. The multiplicative term changes illumination without rewriting base color; the residual accommodates reflections and other effects that this product cannot explain. Sharing albedo is a content-preservation prior, not a claim that real wet or snow-covered surfaces have unchanged reflectance.

Differentiable rasterization projects colors and intrinsic attributes using the same geometry and visibility weights, yielding RGB $\widehat{I}^x$ and intrinsic maps $\widehat{A}^x$, $\widehat{N}^x$, and $\widehat{E}^x$. Hats distinguish rendered maps from offline estimates. Each observation thus provides two paths into the same representation: RGB losses constrain final appearance, while feature losses constrain its intrinsic factors. This explicit decomposition is applied to the background; dynamic objects retain weather-conditioned RGB appearance.

D. Structure-Preserving Weather Refinement

Intrinsic representation alone does not identify every teacher error: an appearance branch could still reproduce a cloud over a building. Stage II therefore controls both *where* teacher content is learned and *how* weather appearance changes. The scene first undergoes a short recovery using original observations to restore a content anchor. During subsequent refinement, geometry, shared features, albedo/normal heads, and the original-weather branch are fixed. Optimization concentrates on target-weather appearance rather than changing the scene to fit teacher artifacts.

1) Structure-Preservation Gate: An external segmenter predicts labels S^o, S^w for the original and teacher images. Semantic disagreement in protected content classes, such as buildings, roads, and vehicles, defines a conflict mask M_s. A second mask M_i detects albedo/normal discrepancies near those conflicts. Semantic evidence identifies changes of object category, while intrinsic evidence extends protection to nearby material or structural inconsistencies. Their combination is dilated and smoothed into a soft gate

$$G = \mathcal{B}(\mathcal{D}(M_s \vee M_i))\,, \qquad G \in [0,1], \tag{2}$$

where $\mathcal{D}$ and $\mathcal{B}$ denote dilation and boundary smoothing. Larger values indicate stronger protection. The gate uses external intrinsic pseudo-maps, not predicted Gaussian attributes; irradiance is excluded because illumination should change with weather. Exact conflict definitions are given in Supplementary Sec. S2.

In protected regions, we transfer teacher color statistics onto the original spatial content. Let $\mathcal{S}_w(I^o)$ match the original image to reliable teacher means and variances, class-wise when sufficient pixels are available. The safe teacher is

$$T^w_{\mathrm{safe}} = (1-G) \odot T^w + G \odot \mathcal{S}_w(I^o). \tag{3}$$

Thus, reliable regions preserve full teacher appearance, while conflicting regions receive its color style without replacing original structures. Simply substituting original RGB would retain the original lighting; statistical transfer instead supplies a weather-compatible tone. A gated edge loss protects local detail, and the soft boundary avoids an abrupt switch between supervision sources.

2) Per-Gaussian Weather Transport: Two zero-initialized scalars, u_i^w and v_i^w, are attached to each background Gaussian. They modulate irradiance and residuals as

$$\begin{aligned} \overline{E}_i^w &= E_i^w \exp(2\tanh u_i^w), \\ \overline{R}_i^w &= R_i^w + 2\tanh(v_i^w)\mathbf{1}_3, \\ \bar{c}_i^w &= \mathrm{clip}\Big(A_i \odot \overline{E}_i^w + \lambda_R \overline{R}_i^w, 0, 1\Big)\,. \end{aligned} \tag{4}$$

Here $\mathbf{1}_3$ broadcasts a scalar across RGB channels. The multiplicative response remains positive, while the bounded additive response adjusts residual intensity. Zero initialization recovers the preceding appearance, avoiding an imposed initial offset. Scalar transport avoids independent per-channel offsets, while target-weather heads and sky are also optimized to express color changes. Because transport is attached to three-dimensional primitives, observations of the same Gaussian share its adaptation across views. It updates scene appearance rather than compositing teacher images at inference.

E. Training Objectives

1) Soft Intrinsic Feature Priors: For an intrinsic map Z, a parameter-free encoder combines pooled values and Sobel edge magnitudes:

$$\Phi(Z; M) = \nu([\mathcal{P}_s(M \odot Z), \mathcal{P}_s(\mathcal{E}_{\mathrm{sob}}(M \odot Z))])\,. \tag{5}$$

M masks sky and ego-vehicle pixels, $\mathcal{P}_s$ is average pooling, brackets denote channel concatenation, and ν normalizes each patch feature. The resulting six-channel representation retains coarse color and structural variation while reducing sensitivity to individual pixels.

Let d_M be cosine distance averaged over patches, weighted by their valid-pixel fractions. Writing $F_Z^x = \Phi(\widehat{Z}^x; M)$ and $\widetilde{F}_Z^x = \Phi(\widetilde{Z}^x; M)$, shared-content supervision is

$$\mathcal{L}_{AN}^x = \tfrac{1}{2} \sum_{Z \in \{A,N\}} d_M(F_Z^x, \widetilde{F}_Z^o). \tag{6}$$

Both original and target-weather predictions thus use the original intrinsic reference. Irradiance instead follows the teacher's change direction:

$$\mathcal{L}_{\mathrm{dir}} = d_M(F_E^w - \widetilde{F}_E^o, \widetilde{F}_E^w - \widetilde{F}_E^o). \tag{7}$$

This separates stable content from weather-dependent illumination: albedo and normals approach a common reference, whereas irradiance is encouraged to change in the teacher's direction. Comparing direction, rather than absolute irradiance, allows variation in the magnitude of the learned response. The encoder introduces no additional trainable network.

2) Stage I: Joint Scene Learning: RGB reconstruction uses $\mathcal{L}_{\mathrm{rgb}}^x = 0.8\,\mathrm{MAE}(\widehat{I}^x, I^x) + 0.2[1 - \mathrm{SSIM}(\widehat{I}^x, I^x)]$, where $I^w = T^w$. The full objective is

$$\mathcal{L}_I = \sum_{x \in \{o,w\}} (\mathcal{L}_{\mathrm{rgb}}^x + \lambda_{AN}\mathcal{L}_{AN}^x) + \lambda_{\mathrm{dir}}\mathcal{L}_{\mathrm{dir}} + \mathcal{L}_{\mathrm{aux}}. \tag{8}$$

TABLE II
QUANTITATIVE COMPARISON ON NUSCENES USING ALL 100 TIMESTAMPS PER SCENE. DATASET-LEVEL RESULTS EQUALLY WEIGHT TEN SCENES. BOLD DENOTES THE BEST VALUE IN EACH COLUMN.

Method	Rain				Snow			
	CLIP-S↑	CLIP-DS↑	Sem-CS↑	$E_{\text{warp}}\downarrow$	CLIP-S↑	CLIP-DS↑	Sem-CS↑	$E_{\text{warp}}\downarrow$
Qwen-Image	0.8248	**0.2090**	0.6748	0.0294	0.7648	0.2692	0.6253	0.0516
FRESCO	0.8070	0.0543	0.5693	0.0174	0.7885	0.0897	0.5443	0.0172
IntrinsicWeather	0.6682	0.1380	0.5497	0.0379	0.6437	0.1563	0.4969	0.0529
WeatherEdit	0.8443	0.1122	0.6863	0.0148	0.8269	0.1412	0.6839	0.0127
WeatherCity	0.8582	0.1289	0.7129	0.0023	0.8652	0.1876	0.6772	0.0038
IGSW (Ours)	**0.8718**	0.1599	**0.8992**	**0.0021**	**0.8782**	**0.2732**	**0.8792**	**0.0028**

$\mathcal{L}_{\text{aux}}$ retains original-image VGG content, occupancy, sparse-depth, camera-affine, shape, and dynamic regularization from the base scene model. RGB fitting establishes visible weather appearance; the intrinsic terms constrain how that appearance is distributed among the decoded factors. Both objectives act through differentiable rendering, rather than learning an independent two-dimensional intrinsic predictor.

3) Stage II: Protected Appearance Learning: The safe teacher supervises both full-resolution and pooled RGB, denoted by $\mathcal{L}_{\text{weather}}$. A local VGG feature-matching loss $\mathcal{L}_{\text{local}}$ tolerates small teacher misalignments, while $\mathcal{L}_{\text{edge}}$ preserves original Sobel edges under G. With $\mathcal{L}_{uv}$ penalizing squared transport magnitudes,

$$\begin{aligned}\mathcal{L}_{II} = \mathcal{L}_{\text{weather}} + \lambda_{\text{local}}\mathcal{L}_{\text{local}} + \lambda_{\text{edge}}\mathcal{L}_{\text{edge}} \\ + \lambda_{\text{dir}}\mathcal{L}_{\text{dir}} + \lambda_{uv}\mathcal{L}_{uv}.\end{aligned} \tag{9}$$

Shared intrinsic parameters remain fixed. Supplementary Secs. S1–S3 detail the losses and settings. Inference requires only the scene, camera, time, and weather.

IV. EXPERIMENTS

A. Experimental Setup

Data and training. We evaluate rain and snow editing on ten nuScenes scenes [25] and on the Waymo Open Dataset [26]. Each scene contains 100 synchronized timestamps from three front-facing cameras: 300 images per weather and 297 within-camera adjacent pairs. Qwen teachers and intrinsic priors are generated offline from the same observations. Separate original/rain and original/snow models are trained at 800×450 for nuScenes and 960×640 for Waymo. Main results use complete sequences without added weather particles.

Stage I runs for 30,000 steps. Stage II uses 800 original-view recovery steps and 3,000 weather-refinement steps. Every fifth timestamp is excluded from weather refinement, while all timestamps contribute to evaluation. This is a post-training split; Stage I uses all observations. Optimizers, loss weights, and gate settings appear in Supplementary Secs. S2–S3.

Metrics. CLIP-S measures image-feature similarity to the original image; CLIP-DS aligns the original-to-weather image change with a fixed text-feature change. Both use CLIP ViT-B/32 [27]. Sem-CS measures frequency-weighted semantic IoU using ADE20K-trained UPerNet–ConvNeXt-XL [29], [30], consistently resizing images to 400×225 before its supplied preprocessing. E_{warp} measures squared RGB differences between adjacent weather frames aligned by RAFT-Large flow from original images. All methods reuse the same flow and valid masks. Scores average all 300 images or 297 frame pairs, with equal scene weighting for nuScenes. Full definitions appear in Supplementary Sec. S4.

Baselines. Table I compares Qwen-Image, FRESCO, IntrinsicWeather, WeatherEdit, WeatherCity, and IGSW. Scene-optimization methods using edited-image supervision share Qwen teachers. WeatherCity provides the direct RGB-based baseline for the intrinsic representation and post-training ablations.

B. Quantitative and Qualitative Results

Tables II and III show that IGSW achieves the highest CLIP-S and Sem-CS and the lowest E_{warp} in all four dataset–weather settings, with the highest CLIP-DS in three. On nuScenes, rain/snow Sem-CS reaches 0.8992/0.8792 and temporal error falls to 0.0021/0.0028. Qwen-Image has higher rain CLIP-DS (0.2090 versus 0.1599), but lower semantic consistency and substantially higher temporal error. IGSW therefore improves the balance between weather change and scene preservation rather than maximizing edit strength alone.

On Waymo, IGSW reaches CLIP-S of 0.9031/0.8971 and CLIP-DS of 0.2940/0.2869 for rain/snow. The accompanying Sem-CS and temporal improvements indicate that stronger weather appearance need not come at the expense of stable scene content. Compared with WeatherEdit, its strongest CLIP-S baseline on Waymo, IGSW also improves semantic consistency under both conditions.

Figures 3 and 4 compare matched views. In these examples, FRESCO and WeatherEdit express relatively weak weather changes, while IntrinsicWeather exhibits local artifacts. WeatherCity preserves scene-level continuity but can inherit teacher errors and muted weather appearance. IGSW more closely reproduces the teacher's illumination and atmosphere while retaining buildings, roads, and traffic participants.

C. Temporal and Structural Consistency

Table IV reports nuScenes temporal errors. IGSW reduces E_{warp} by more than an order of magnitude relative to the Qwen teacher and also improves upon WeatherCity. Figure 5 illustrates rain appearance at five matched timestamps. The shared scene representation stabilizes changes that vary across independently edited images, including illumination and road

TABLE III
QUANTITATIVE COMPARISON ON THE WAYMO OPEN DATASET USING COMPLETE 100-TIMESTAMP SEQUENCES. BOLD DENOTES THE BEST VALUE IN EACH COLUMN.

Method	Rain				Snow			
	CLIP-S↑	CLIP-DS↑	Sem-CS↑	E_{warp} ↓	CLIP-S↑	CLIP-DS↑	Sem-CS↑	E_{warp} ↓
Qwen-Image	0.8450	0.1942	0.5099	0.0547	0.7971	0.2007	0.4571	0.0672
FRESCO	0.8703	0.1487	0.6376	0.0195	0.8258	0.1557	0.6710	0.0184
IntrinsicWeather	0.6046	0.1601	0.3919	0.0224	0.7317	0.1929	0.5676	0.0273
WeatherEdit	0.9015	0.2095	0.8036	0.0118	0.8532	0.2055	0.8025	0.0106
WeatherCity	0.7773	0.1375	0.7398	0.0090	0.8002	0.0937	0.6633	0.0101
IGSW (Ours)	**0.9031**	**0.2940**	**0.8723**	**0.0009**	**0.8971**	**0.2869**	**0.8050**	**0.0019**



Fig. 3. Qualitative rain and snow comparisons on nuScenes. Rows show original observations, Qwen-Image, FRESCO, IntrinsicWeather, WeatherEdit, WeatherCity, and IGSW. Columns use corresponding views and weather conditions.

TABLE IV
TEMPORAL WARPING ERROR ON NUSCENES. LOWER IS BETTER.

Method	Rain E_{warp} ↓	Snow E_{warp} ↓
Qwen-Image	0.0294	0.0516
FRESCO	0.0174	0.0172
IntrinsicWeather	0.0379	0.0529
WeatherEdit	0.0148	0.0127
WeatherCity	0.0023	0.0038
IGSW (Ours)	**0.0021**	**0.0028**

appearance. Supplementary Secs. S5–S6 provide nuScenes snow and Waymo rain/snow sequences at the same fixed timestamps.

Temporal stability alone does not describe editing quality: a weak or overly smooth edit can also give low warping error. The comparisons therefore pair full-sequence temporal scores with weather-direction scores and visible scene detail. In the displayed Waymo results, IGSW expresses stronger weather appearance than the baseline while maintaining lower motion-aligned error.

Figure 6 isolates a structural failure: clouds in the Qwen teacher cover part of a building that should remain visible. WeatherCity transfers some of this error. IGSW instead retains the building layout while adapting local color and illumination. This illustrates the purpose of protected supervision: teacher

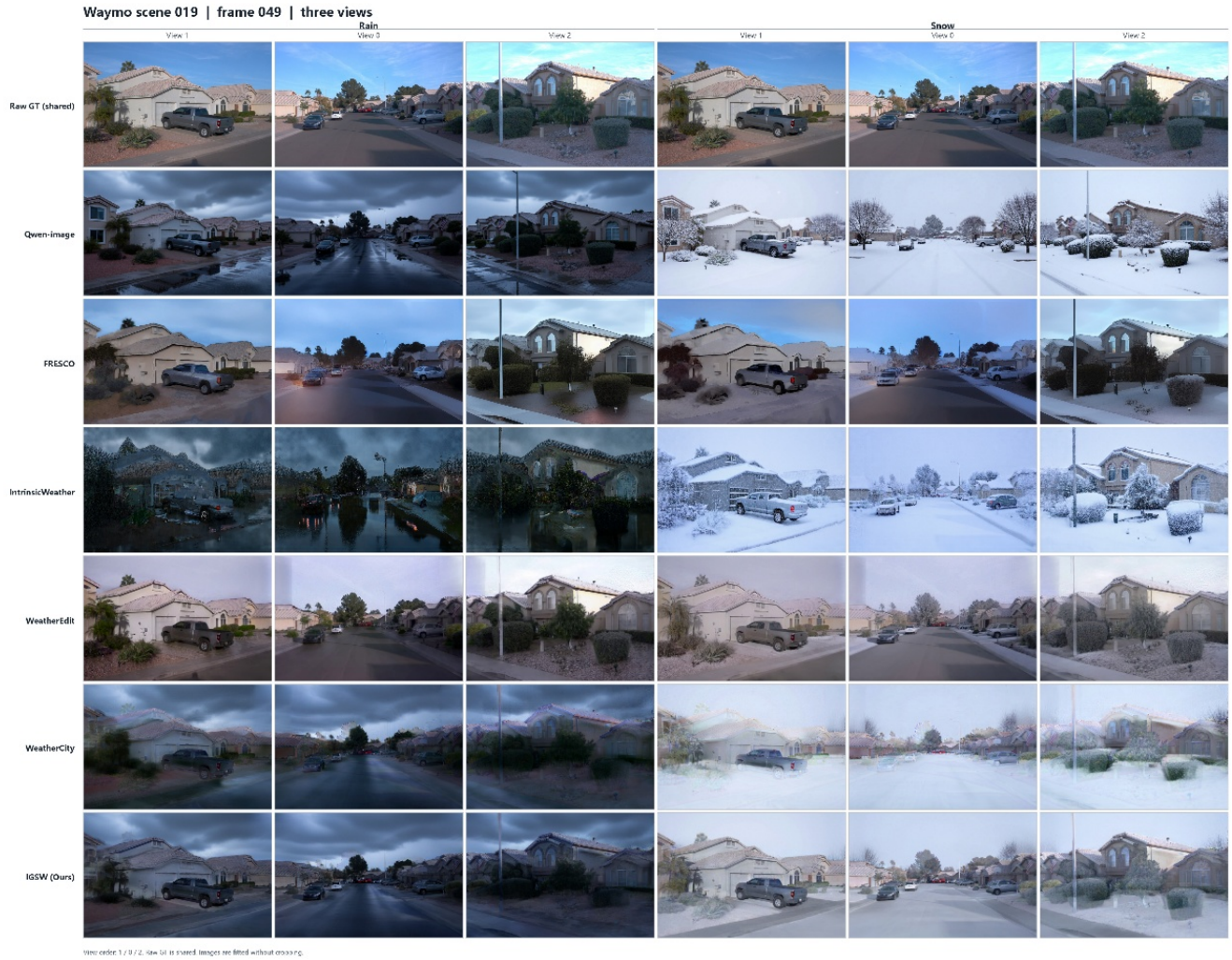


Fig. 4. Qualitative rain and snow comparisons on Waymo. Method order matches Fig. 3; viewpoints and weather conditions are aligned across methods.

style is learned without requiring the scene to reproduce every teacher region.

The distinction is important when interpreting a close visual match to the teacher. In this example, copying the teacher more faithfully would reproduce the missing building, even though the image might appear convincingly rainy. The structure gate redirects supervision at the conflict, while reliable road and sky regions continue to receive teacher appearance. The three views show how structure preservation and weather transfer can coexist within a shared scene rather than being independent per-image objectives.

The temporal and structural comparisons address different aspects of this behavior. Figure 5 examines appearance over time; Fig. 6 examines retained content across cameras. The corresponding full-sequence metrics quantify temporal stability and semantic agreement, while the displayed details make their visual implications easier to inspect. The ablations below separate the contributions of the intrinsic representation and the refinement mechanism.

D. Ablation Studies

1) Module Ablations: We compare WeatherCity, Stage I only, Stage II only, and the full two-stage model using complete 100-timestamp evaluation. Stage I is trained for 30,000 steps; the Stage II-only variant applies protected refinement without intrinsic Stage I modeling. This separates a structured appearance representation from subsequent appearance optimization.

Table V shows modest changes from Stage I alone, consistent with its visually similar appearance to the baseline in Fig. 7. Stage II alone improves teacher-style transfer and gives the lowest temporal error. Combining both stages yields the highest CLIP-S, CLIP-DS, and Sem-CS, with slightly higher temporal error than Stage II alone but lower error than the baseline. Intrinsic modeling and refinement therefore provide complementary content and appearance benefits.

2) Component Ablations: The four components are (a) shared albedo/normals, (b) weather-specific irradiance/residuals, (c) the structure-preservation gate, and (d) Gaussian transport. Components (a) and (b) are ablated during Stage I training; (c) and (d) are ablated during refinement. Table VI and Fig. 8 compare their removal with the baseline and full model.

Removing shared albedo/normals (a) lowers CLIP-S, CLIP-

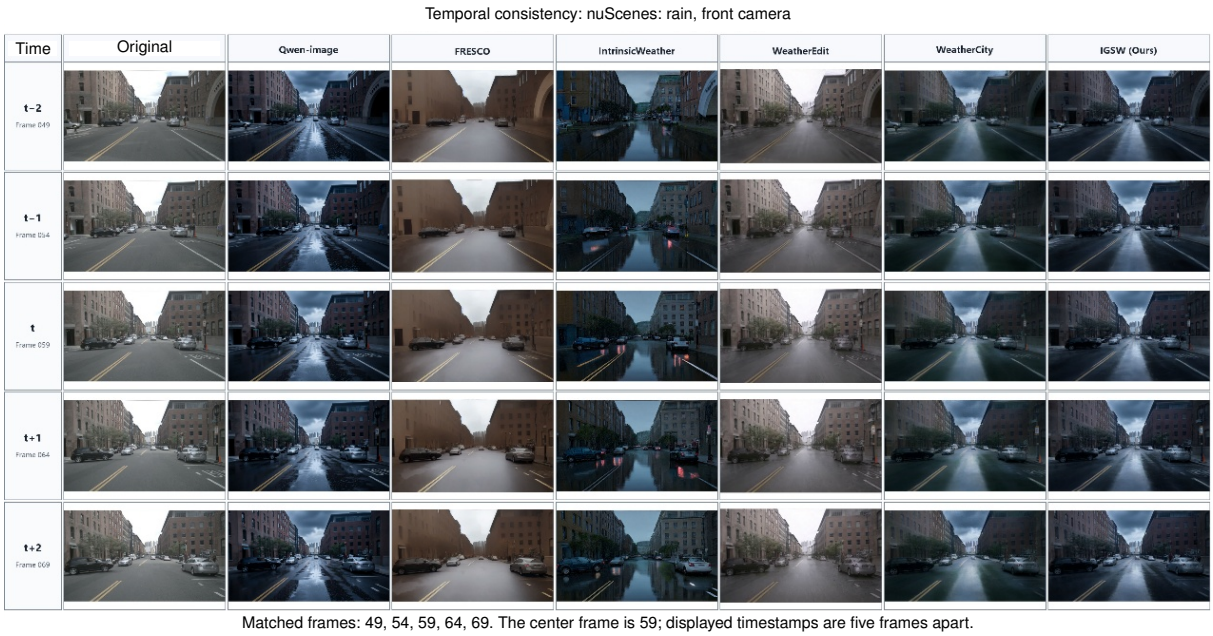


Fig. 5. Temporal rain comparison on nuScenes at frames 49, 54, 59, 64, and 69. These fixed samples illustrate appearance evolution; quantitative evaluation uses all 297 adjacent frame pairs. Snow and Waymo sequences are in the supplementary material.

TABLE V
MODULE ABLATIONS WITH COMPLETE 100-TIMESTAMP, THREE-VIEW EVALUATION. BEST VALUES ARE BOLD.

Method	Rain				Snow			
	CLIP-S↑	CLIP-DS↑	Sem-CS↑	E_{warp} ↓	CLIP-S↑	CLIP-DS↑	Sem-CS↑	E_{warp} ↓
WeatherCity (baseline)	0.858270	0.128999	0.712923	0.002399	0.865202	0.187677	0.677263	0.003811
Stage I only	0.858933	0.132594	0.714228	0.002265	0.866956	0.188946	0.677516	0.003186
Stage II only	0.866129	0.127921	0.898980	**0.001632**	0.864229	0.220130	0.877846	**0.002207**
IGSW (Stage I + II)	**0.87186**	**0.159925**	**0.89925**	0.002152	**0.878254**	**0.273236**	**0.879220**	0.002844



Fig. 6. Structural consistency at a fixed timestamp and three views. Rows: original, Qwen-Image, WeatherCity, and IGSW. Teacher clouds incorrectly cover a building; IGSW retains more of its structure while adapting rainy appearance.

DS, and Sem-CS in both weather conditions. The shared representation therefore contributes to maintaining content during weather adaptation. Temporal error becomes slightly lower, showing that shared content is not simply a mechanism for minimizing frame-to-frame variation.

Removing weather-specific irradiance/residuals (b) raises rain CLIP-S but weakens weather-direction and semantic scores. Preserving original-like appearance is easier when the weather response is restricted, yet target-weather expression suffers. The displayed ablation correspondingly shows weaker illumination and atmosphere changes.

Without the structure gate (c), CLIP-S and Sem-CS decrease and temporal error rises in both conditions. Snow CLIP-DS increases to 0.280840, but the accompanying semantic loss demonstrates that a stronger edit can transfer more teacher errors. The gate is thus important for determining which appearance changes should be accepted, rather than merely constraining their overall magnitude.

Without transport (d), temporal error is lowest and snow CLIP-S is highest, while weather-direction scores decrease. Its illumination remains closer to the baseline in the shown examples. Point-wise transport strengthens local weather responses instead of optimizing smoothness alone. Together, these ablations support distinct roles for intrinsic content modeling, weather-specific decoding, protected supervision, and local adaptation. The complete model improves all four metrics over the baseline while balancing the trade-offs exposed by



Fig. 7. Module ablations for rain (left) and snow (right). Rows: original, Qwen teacher, WeatherCity, Stage I only, Stage II only, and full IGSW. Each panel contains three matched camera views.

TABLE VI
COMPONENT ABLATIONS WITH COMPLETE 100-TIMESTAMP, THREE-VIEW EVALUATION. COMPONENTS: SHARED ALBEDO/NORMALS (A), WEATHER-SPECIFIC IRRADIANCE/RESIDUALS (B), STRUCTURE-PRESERVATION GATE (C), AND WEATHER TRANSPORT (D). BEST VALUES ARE BOLD.

Method	Rain				Snow			
	CLIP-S↑	CLIP-DS↑	Sem-CS↑	$E_{\text{warp}}\downarrow$	CLIP-S↑	CLIP-DS↑	Sem-CS↑	$E_{\text{warp}}\downarrow$
WeatherCity (baseline)	0.858270	0.128999	0.712923	0.002399	0.865202	0.187677	0.677263	0.003811
w/o (a): $b+c+d$	0.861298	0.140355	0.888302	0.002114	0.849778	0.266639	0.871814	0.002821
w/o (b): $a+c+d$	**0.876341**	0.136436	0.892923	0.002052	0.861993	0.262443	0.878149	0.002831
w/o (c): $a+b+d$	0.847179	0.155122	0.861325	0.002178	0.838087	**0.280840**	0.794376	0.003027
w/o (d): $a+b+c$	0.851419	0.129318	0.882965	**0.001655**	**0.882579**	0.191582	0.871025	**0.002463**
IGSW: $a+b+c+d$	0.87186	**0.159925**	**0.89925**	0.002152	0.878254	0.273236	**0.879220**	0.002844

individual removals.

V. DISCUSSION AND CONCLUSION

IGSW combines intrinsic Gaussian modeling with structure-preserving weather post-training. Shared content and weather-specific appearance are learned using soft intrinsic priors, then refined through reliable teacher supervision and point-attached transport. The resulting scene better balances weather expression, semantic preservation, and temporal stability across the reported rain and snow experiments.

The method still depends on the quality of image editing, intrinsic estimation, and semantic segmentation. Strong occlusion or severe teacher distortion can defeat the structure gate, and shared albedo simplifies weather-induced material changes. The experiments assess matched-view weather editing rather than fully unseen-time reconstruction or downstream driving performance. Future work will address richer surface responses and reliable transfer across broader weather conditions.

REFERENCES

[1] B. Kerbl, G. Kopanas, T. Leimkuehler, and G. Drettakis, "3D Gaussian splatting for real-time radiance field rendering," ACM Transactions on Graphics, vol. 42, no. 4, pp. 1–14, 2023.

[2] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R.Ramamoorthi, and R. Ng, "NeRF: Representing scenes as neural radiance fields for view synthesis," in Proc. ECCV, 2020, pp. 405–421.

[3] Y. Yan, H. Lin, C. Zhou, W. Wang, H. Sun, K. Zhan, X. Lang, X. Zhou, and S. Peng, "Street Gaussians: Modeling dynamic urban scenes with Gaussian splatting," arXiv:2401.01339, 2024.

[4] X. Zhou, Z. Lin, X. Shan, Y. Wang, D. Sun, and M.-H. Yang, "DrivingGaussian: Composite Gaussian splatting for surrounding dynamic autonomous driving scenes," in Proc. CVPR, 2024.

[5] W. Wu, H. Guan, Z. Liu, and H. Wang, "WeatherCity: Urban scene reconstruction with controllable multi-weather transformation," arXiv:2602.22096, 2026.

[6] "WeatherEdit: Controllable weather editing with 4D Gaussian field," arXiv:2505.20471, 2025.

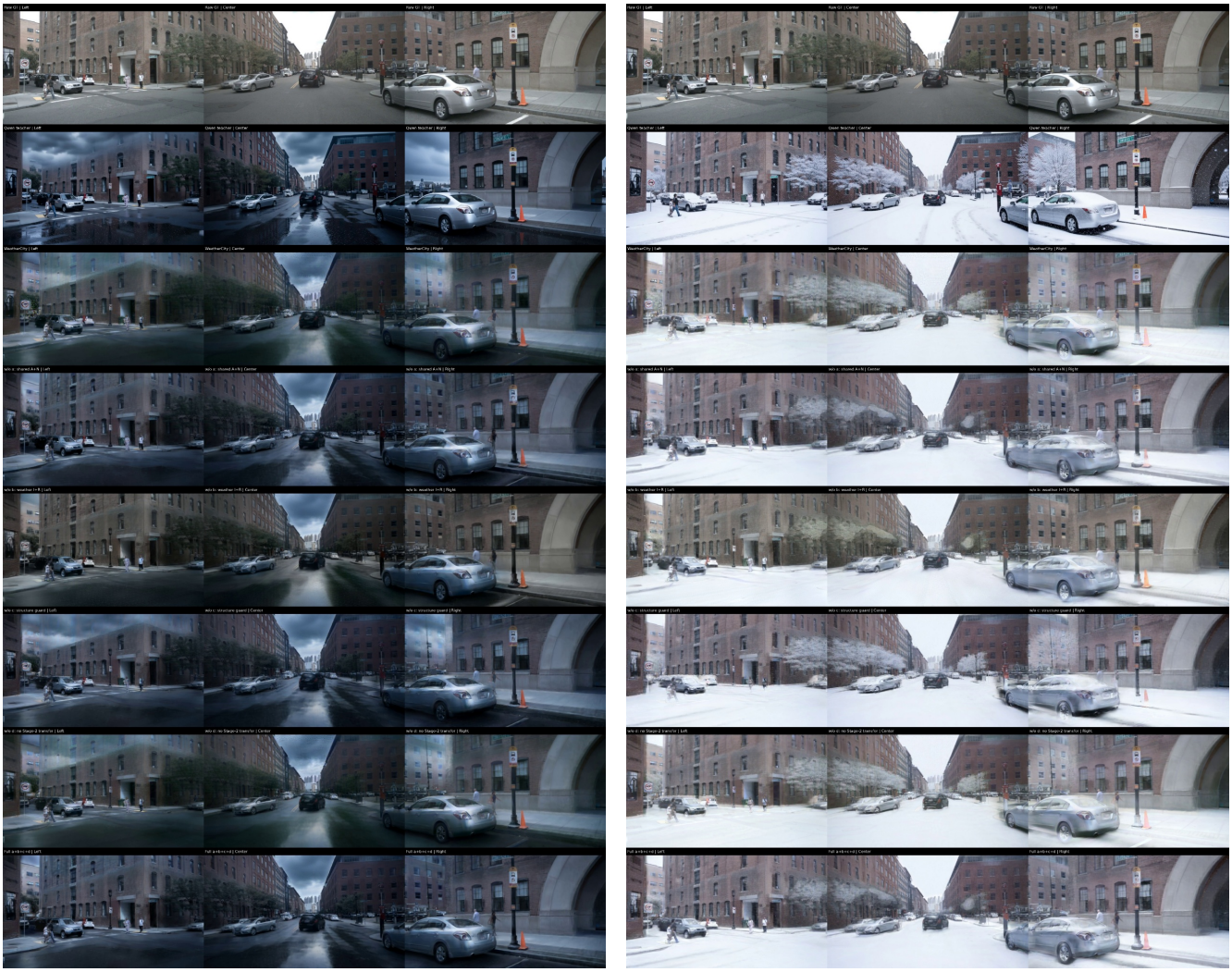


Fig. 8. Component ablations for rain (left) and snow (right). Rows: original, Qwen teacher, WeatherCity, w/o (a), w/o (b), w/o (c), w/o (d), and full IGSW. Components are defined in Table VI. Each panel contains three matched views.

[7] "Weather-Magician: Reconstruction and rendering framework for 4D weather synthesis in real time," arXiv:2505.19919, 2025.

[8] "RainyGS: Efficient rain synthesis with physically-based Gaussian splatting," 2025.

[9] "Let It Snow! Animating static Gaussian scenes with dynamic weather effects," arXiv:2504.05296, 2025.

[10] Z. Li, L. Wang, X. Zhang, and others, "ClimateNeRF: Extreme weather synthesis in neural radiance field," in Proc. ICCV, 2023.

[11] "WeatherGAN: Multi-domain weather translation using generative adversarial networks," 2021.

[12] J. Valanarasu, R. Yasarla, and V. Patel, "TransWeather: Transformer-based restoration of images degraded by adverse weather conditions," arXiv:2111.14813, 2021.

[13] T. Brooks, A. Holynski, and A. A. Efros, "InstructPix2Pix: Learning to follow image editing instructions," in Proc. CVPR, 2023.

[14] L. Zhang, A. Rao, and M. Agrawala, "Adding conditional control to text-to-image diffusion models," in Proc. ICCV, 2023.

[15] A. Hertz, R. Mokady, J. Tenenbaum, K. Aberman, Y. Pritch, and D. Cohen-Or, "Prompt-to-Prompt image editing with cross-attention control," arXiv:2208.01626, 2022.

[16] M. Geyer, O. Bar-Tal, S. Bagon, and T. Dekel, "TokenFlow: Consistent diffusion features for consistent video editing," arXiv:2307.10373, 2023.

[17] S. Yang, Y. Zhou, Z. Liu, and C. C. Loy, "FRESCO: Spatial-temporal correspondence for zero-shot video translation," in Proc. CVPR, 2024.

[18] Qwen Team, "Qwen-Image technical report," arXiv:2508.02324, 2025.

[19] Y. Zhu, Z.-L. Zhu, J. Yang, M. Hasan, J. Xie, and B. Wang, "IntrinsicWeather: Controllable weather editing in intrinsic space," arXiv:2508.06982, 2026.

[20] D. Kocsis, V. Sitzmann, and M. Nießner, "Intrinsic image diffusion for indoor single-view material estimation," in Proc. CVPR, 2024.

[21] "IntrinsicEdit: Precise generative image manipulation in intrinsic space," ACM Transactions on Graphics, 2025.

[22] "RGB↔X: Image decomposition and synthesis using material- and lighting-aware diffusion models," ACM Transactions on Graphics, 2024.

[23] "Uni-Renderer: Unifying rendering and inverse rendering via dual stream diffusion," arXiv:2412.15050, 2024.

[24] "DiffusionRenderer: Neural inverse and forward rendering with video diffusion models," arXiv:2501.18590, 2025.

[25] H. Caesar et al., "nuScenes: A multimodal dataset for autonomous driving," in Proc. CVPR, 2020.

[26] P. Sun et al., "Scalability in perception for autonomous driving: Waymo Open Dataset," in Proc. CVPR, 2020.

[27] A. Radford et al., "Learning transferable visual models from natural language supervision," in Proc. ICML, 2021.

[28] J. Hessel, A. Holtzman, M. Forbes, R. Le Bras, and Y. Choi, "CLIPScore: A reference-free evaluation metric for image captioning," in Proc. EMNLP, 2021.

[29] B. Zhou, H. Zhao, X. Puig, S. Fidler, A. Barriuso, and A. Torralba, "Scene parsing through ADE20K dataset," in Proc. CVPR, 2017.

[30] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in Proc. CVPR, 2022.

Supplementary Material

IGSW: Multi-Weather Urban Scene Reconstruction with Intrinsic-Decoupled Gaussian Splatting

S1. Representation and Intrinsic Feature Supervision

A. Notation and Data Flow

The method distinguishes attributes attached to Gaussians, maps rendered from those attributes, and external inverse-rendering estimates. The notation below summarizes these roles. Superscript o denotes the original condition and w a target weather; $x \in \{o, w\}$ selects either. Subscripts t, v identify a synchronized timestamp and camera. We omit them when discussing one training observation.

Symbol	Meaning
I^o, T^w	Original RGB and edited weather teacher
$f_i \in \mathbb{R}^{32}$	Learned feature of background Gaussian i
$A_i, N_i \in \mathbb{R}^3$	Shared albedo and normal attributes
$E_i^x, R_i^x \in \mathbb{R}^3$	Weather-specific irradiance and residual
$\widehat{Z}^x$	Rendered map for attribute Z
$\widetilde{Z}^x$	Fixed external intrinsic pseudo-map
$\Phi(Z; M)$	Masked, normalized patch features
$G \in [0, 1]$	Image-space structure-protection gate
$u_i^w, v_i^w \in \mathbb{R}$	Gaussian-attached transport scalars

The latent code f_i is optimized through rendering losses. Its coordinates have no prescribed correspondence to material or illumination values; semantic roles are introduced by the decoder heads and their supervision. Shared heads receive the same feature under either weather. Weather-specific heads use separate parameters but draw on this common scene code. Geometry determines projection and visibility, while decoded attributes determine the quantities accumulated by rasterization.

The normal head provides an auxiliary learned structural cue, not a covariance-derived surface normal. It is converted to camera coordinates and the pseudo-map value range before feature comparison. Normals do not enter the RGB product directly. The same rendering geometry links normal supervision to the scene, but an accurate normal feature does not independently certify accurate three-dimensional geometry.

B. Parameter-Free Patch Encoder

For a three-channel intrinsic map Z, let M be the binary valid-pixel mask after excluding sky and ego-vehicle regions. The Sobel magnitude is computed channel-wise:

$$\mathcal{E}_{\mathrm{sob}}(Z) = \sqrt{(K_x * Z)^2 + (K_y * Z)^2 + \epsilon}, \tag{S1}$$

where K_x, K_y are horizontal and vertical Sobel kernels, $*$ denotes convolution, and ϵ stabilizes numerical operations. The encoder concatenates pooled values and pooled edge magnitudes, then normalizes the six-channel vector at each patch. Average pooling uses scale $s = 16$. The three value channels retain coarse intrinsic appearance, and the three edge channels retain local structural variation. Neither metallicity nor roughness contributes to this encoder or its losses.

Pooling reduces the influence of isolated pixel errors without discarding all spatial organization. The edge branch complements average values: patches with similar mean color can still differ in their boundaries. Normalization makes the subsequent comparison emphasize feature direction. This lightweight representation is fixed throughout optimization; no additional feature network is learned jointly with the Gaussians.

C. Masked Feature Distance

Let $a_p, b_p \in \mathbb{R}^6$ denote features at patch p, $\omega_p = \mathcal{P}_s(M)_p$ its valid-pixel fraction, and $\mathcal{V} = \{p : \omega_p > 10^{-4}\}$. For nonempty $\mathcal{V}$, the distance is

$$d_M(a, b) = \frac{\sum_{p \in \mathcal{V}} \omega_p \left[1 - \frac{a_p^\top b_p}{\max(\|a_p\|_2, \epsilon) \max(\|b_p\|_2, \epsilon)}\right]}{\sum_{p \in \mathcal{V}} \omega_p + \epsilon}. \tag{S2}$$

Valid patches contribute in proportion to their support. The same mask is applied before extracting rendered and pseudo-map features. Small constants stabilize normalization and weighting. If no patch passes the support threshold, the implementation falls back to the unweighted mean feature distance.

Albedo and normal losses compare $\Phi(\widehat{Z}^x; M)$ with $\Phi(\widetilde{Z}^o; M)$ for both original and target weather. Their common reference discourages weather-dependent changes to stable content. Irradiance instead compares the predicted and teacher feature changes relative to $\Phi(\widetilde{E}^o; M)$. Differences are taken after encoding, rather than encoding an RGB or irradiance difference image. A cosine distance aligns the direction of these feature changes without specifying their exact magnitude.

D. Relation to RGB Supervision

Intrinsic losses supplement the reconstruction objective rather than replace it. RGB loss determines whether the composed color reproduces the observed original or edited teacher, while intrinsic losses influence how this appearance is represented. All rendered quantities use the same Gaussian geometry and visibility. Their gradients therefore meet in the scene parameters and latent features. This differs from training an independent two-dimensional inverse renderer whose predictions would not constrain the scene representation.

During Stage II, shared content is fixed and the irradiance-direction term remains active for weather appearance. Shared albedo/normal features are not reoptimized under a nominally active but gradient-free objective. The gate described next uses external pseudo-maps, so teacher-reliability estimation remains distinct from the trainable Gaussian heads.

S2. Structure Gate and Refinement Losses

A. Semantic and Intrinsic Conflicts

Let P indicate protected classes in the original semantic prediction. We use stable scene categories including buildings, walls, roads, sidewalks, pedestrians, vehicles, houses, and skyscrapers. With disagreement $M_\Delta = \mathbf{1}[S^o \neq S^w]$, the semantic mask is $M_s = P \odot M_\Delta$.

The intrinsic discrepancy combines albedo and normal differences:

$$D = \tfrac{1}{2}\left[\frac{\|\widetilde{A}^o - \widetilde{A}^w\|_1}{3} + \frac{1 - \widetilde{n}^{o\top}\widetilde{n}^w}{2}\right], \tag{S3}$$

where $\widetilde{n}^o, \widetilde{n}^w$ are unit normals decoded from pseudo-maps. Intrinsic conflicts are restricted to semantic-disagreement neighborhoods:

$$M_i = P \odot \mathcal{D}_9(M_\Delta) \odot \mathbf{1}[D > \tau]. \tag{S4}$$

The final gate is $G = \mathcal{B}_{19}(\mathcal{D}_{11}(M_s \vee M_i))$, where subscripts specify square kernel widths and $\tau = 0.24$. It depends on external original/teacher evidence, so its construction does not require the learned albedo/normal heads. Excluding irradiance allows genuine illumination changes to remain learnable.

B. Color Transfer in Protected Regions

For semantic class ℓ, estimate channel-wise means m_o^ℓ, m_w^ℓ and variances v_o^ℓ, v_w^ℓ from reliable original and teacher pixels. The bounded gain and transferred image are

$$\begin{aligned} a_w^\ell &= \mathrm{clip}\left(\sqrt{\frac{v_w^\ell + \epsilon}{v_o^\ell + \epsilon}}, 0.65, 1.35\right), \\ \mathcal{S}_w(I^o) &= \mathrm{clip}\left[(I^o - m_o^\ell) \odot a_w^\ell + m_w^\ell, 0, 1\right]. \end{aligned} \tag{S5}$$

Class-wise statistics require at least 256 reliable pixels. Insufficient support triggers a global reliable-pixel fallback, followed by unconflicted-image statistics when needed. The affine color transform retains original spatial content, while the soft gate blends it with the unmodified teacher outside conflicts. This construction is used as a training target, not as a rendered-image post-processing step.

C. Refinement Losses

Writing J for the weather rendering and T for the safe teacher, the two-scale reconstruction loss is

$$\mathcal{L}_{\mathrm{weather}} = 0.3\,\mathrm{MAE}(J, T) + 0.7\,\mathrm{MAE}(\mathcal{P}_8 J, \mathcal{P}_8 T). \tag{S6}$$

The low-frequency term emphasizes weather tone, while the full-resolution term retains local appearance. Local perceptual matching uses frozen VGG features V_ℓ:

$$\begin{aligned} \mathcal{L}_{\mathrm{local}} &= \frac{1}{|\mathcal{L}|}\sum_{\ell\in\mathcal{L}}\frac{1}{|\Omega_\ell|}\sum_{p\in\Omega_\ell}\min_{q\in\mathcal{N}_5(p)} d_\ell(p,q), \\ d_\ell(p,q) &= 1 - \cos(V_\ell(J)_p, V_\ell(T)_q). \end{aligned} \tag{S7}$$

$\mathcal{L} = \{\texttt{relu3_1}, \texttt{relu4_1}\}$, Ω_ℓ is a feature grid, and $\mathcal{N}_5(p)$ is its 5×5 neighborhood. Images are resized to width 400 with preserved aspect ratio. Neighborhood matching tolerates small teacher displacements while retaining appearance supervision.

Protected detail and transport magnitude are constrained by

$$\begin{aligned} \mathcal{L}_{\mathrm{edge}} &= \frac{\sum_p G_p \|\nabla J_p - \nabla I_p^o\|_1}{3\sum_p G_p + \epsilon}, \\ \mathcal{L}_{uv} &= \frac{1}{2N_B}\sum_{i=1}^{N_B}[(u_i^w)^2 + (v_i^w)^2], \end{aligned} \tag{S8}$$

where ∇ is the Sobel operator and N_B the number of background Gaussians. Edge supervision is confined to protected regions; it does not require all weather-image edges to match the original. Transport regularization limits unnecessary local intensity corrections. Together with the retained irradiance-direction prior, these losses separate weather fitting from content protection.

D. Gradient Paths During Refinement

The gate, class-wise color statistics, and safe teacher are fixed inputs to weather refinement. They are constructed from the aligned original/teacher observations, so their values do not change in response to the rendering being optimized. The weather and local perceptual losses backpropagate through the rendered RGB to trainable target-weather appearance. The gated edge term follows the same route but compares protected detail with the original observation. The direction prior acts on rendered irradiance, and transport regularization acts directly on u_i^w, v_i^w.

Geometry and shared content are held fixed during this optimization. Consequently, preserving an original building does not depend on moving Gaussian centers to accommodate the edited cloud. The optimizer instead adjusts allowed appearance variables under the corrected supervision. This distinction also separates the image-space reliability gate from the Gaussian-space transport field: the former determines the target, whereas the latter determines how the scene can respond.

E. Transport Responses

The two scalar transport variables have different effects. The exponential factor changes the positive irradiance response multiplicatively, and the bounded offset changes the residual additively. Their zero initialization leaves both responses unchanged. Subsequent updates remain shared across observations of the same Gaussian, unlike independently optimized image corrections. The magnitude penalty discourages unnecessary changes rather than forcing either response to zero throughout training.

TABLE S1
QWEN-IMAGE TEACHER-GENERATION PROMPTS, TRANSLATED FROM CHINESE FOR PRESENTATION.

Weather	Prompt
Rain	Strictly preserve the original composition, all scene content (including the ground, buildings, plants, vehicles, background, and pedestrians), their positions, and the original artistic style. Convert the scene to rainy weather. Requirements: the image should be clear and realistic; the sky must be overcast with dark clouds; the ground should be wet with puddles and reflections; lighting should be dim and the overall tone cool to create a rainy atmosphere. Do not include clear weather, blue sky and white clouds, sunlight, dry ground, any elements or trees absent from the original image, visual distortions, malformed objects, or incorrect proportions.
Snow	Strictly preserve the original composition, all scene content (including the ground, buildings, vegetation, vehicles, background, and pedestrians), their positions, and the original artistic style. Convert the scene to snowy weather. Requirements: the image should be clear and realistic; the sky should be overcast with falling snow; the ground should be naturally covered with snow; do not add extra vegetation; lighting should be soft and the overall tone cool. If the original image contains green leaves, turn them into snow-covered bare branches. Do not include clear weather, warm tones, sunlight, melting snow, elements unrelated to or absent from the original image, visual distortions, deformations, or incorrect proportions.

S3. IMPLEMENTATION AND TEACHER PROMPTS

A. Data Preparation and Scene Learning

nuScenes images are loaded from 1600×900 at 800×450; Waymo images are loaded from 1920×1280 at 960×640. Each original/rain or original/snow pair forms a separate scene model. RGB images, camera poses, timestamps, LiDAR, masks, and dynamic annotations retain their paired indexing. Qwen teachers and intrinsic pseudo-maps are generated once, then fixed throughout training.

Background appearance features have 32 channels. Each intrinsic MLP has two hidden layers of width 32 and a three-channel output. The residual scale is 0.1; the feature-encoder pooling scale is 16. Each Stage I step samples a timestamp and camera, evaluating original and target-weather supervision for the same observation. Progressive resolution, densification, and pruning follow the base scene model.

Adam learning rates are 2.5×10^{-3} for shared appearance features, 5×10^{-4} for intrinsic heads, and 10^{-4} for weather RGB heads. Position learning rate decays from 1.6×10^{-4} to 1.6×10^{-6} with scene-scale adjustment. Scale, rotation, and opacity use 5×10^{-3}, 10^{-3}, and 5×10^{-2}.

RGB and SSIM weights are 0.8/0.2. Shared albedo/normal feature consistency and irradiance-direction weights are 0.02/0.01. Hard pixel-wise intrinsic losses and confidence weighting are disabled. Original-image content uses frozen VGG-19 `relu4_2` with weight 1. Occupancy, sparse depth, and affine regularization weights are 0.05, 0.01, and 10^{-5}; shape and trajectory regularizers follow the base model.

B. Post-Training and Parameter Protection

Stage I uses 30,000 steps. An 800-step recovery receives only original observations, followed by 3,000 protected weather-refinement steps. During refinement, geometry, shared appearance features, albedo/normal heads, original-weather appearance, dynamic-object state, and camera parameters are fixed. Target-weather heads and sky remain trainable. Point-wise transport variables are zero-initialized and optimized at 3×10^{-3}; other trainable weather parameters use 10^{-4}. Densification and splitting are disabled.

Refinement weights are $\lambda_{\text{local}} = 0.2$, $\lambda_{\text{edge}} = 0.08$, $\lambda_{\text{dir}} = 0.01$, and $\lambda_{uv} = 5 \times 10^{-4}$. Shared albedo/normal losses are not reoptimized when their parameters are fixed. The structure gate and safe-teacher targets are prepared from aligned observations before refinement.

C. Teacher Generation

Table S1 records the teacher-generation prompts. They specify both target weather and content preservation, but the requests do not guarantee geometrically faithful edits. This motivates the gate rather than altering the prompts after inspecting evaluation scores. The prompt text below is an English translation of the Chinese generation instructions; it is distinct from the short metric prompts in Sec. S4.

D. Temporal Split and Inference

Every fifth timestamp is excluded from Stage II weather refinement, leaving 80 optimization and 20 diagnostic timestamps with a common split across cameras. Stage I has already seen all 100 timestamps. The split therefore measures post-training fitting behavior, not unseen-time reconstruction. Main tables always aggregate the complete sequences.

Inference rasterizes the learned scene at a requested camera, time, and weather. The image editor, inverse renderer, segmenter, and VGG networks are training-only components. Reported main comparisons use no additional raindrop or snow-particle pass.

E. Parameter Groups During Refinement

After original-only recovery, weather refinement uses the following parameter groups:

Parameter group	State
Centers, scales, rotations, opacities	Fixed
Shared features and albedo/normal heads	Fixed
Original-weather branch and cameras	Fixed
Target-weather appearance and sky	Trainable
Per-Gaussian scalars u_i^w, v_i^w	Trainable

S4. Fixed Full-Sequence Evaluation

A. CLIP Content and Direction

Let $e(X) = \phi(X)/\max(\|\phi(X)\|_2, \epsilon)$ be the normalized CLIP image embedding and $h(s)$ its normalized text embedding. For an original image I^o and an evaluated weather image J^w,

$$\text{CLIP-S} = e(I^o)^\top e(J^w). \tag{S9}$$

This is image–image similarity, not image-caption CLIPScore. With $\Delta e = e(J^w) - e(I^o)$ and $\Delta h = h(s_w) - h(s_o)$,

$$\text{CLIP-DS} = \frac{\Delta e^\top \Delta h}{\max(\|\Delta e\|_2, \epsilon)\max(\|\Delta h\|_2, \epsilon)}. \tag{S10}$$

The fixed descriptions are "a photo of a street scene" for s_o, and "a photo of a street scene in rainy weather" or "a photo of a street scene in snowy weather" for s_w. The text difference, CLIP ViT-B/32 weights, and supplied image preprocessing are shared by all methods.

B. Semantic Consistency

UPerNet–ConvNeXt-XL predicts ADE20K labels. If n_{cd} counts pixels labeled c in the original and d in the edited image, let $n_c = \sum_d n_{cd}$ and $m_c = \sum_d n_{dc}$. Frequency-weighted IoU is

$$\text{Sem-CS} = \sum_{c:n_c>0} \frac{n_c}{\sum_d n_d} \frac{n_{cc}}{n_c + m_c - n_{cc}}. \tag{S11}$$

Images are resized to 400×225, processed by the supplied 512×512 pipeline, and logits interpolated back before label assignment. Resolution is fixed across methods. The metric measures predicted semantic agreement, not geometric ground truth. As the gate uses the same segmenter, other metrics and visual evidence provide complementary checks.

C. Motion-Aligned Temporal Error

RAFT-Large estimates bidirectional flow from original frames. Forward–backward consistency and image bounds define mask $M^o_{t,v}$. All methods reuse this flow and mask:

$$E^{t,v}_{\text{warp}} = \frac{\sum_p M^o_{t,v}(p)\|d_{t,v}(p)\|_2^2}{\sum_p M^o_{t,v}(p)}, \qquad d_{t,v} = J^w_{t,v} - \mathcal{W}(J^w_{t+1,v}, F^o_{t\to t+1,v}). \tag{S12}$$

$\mathcal{W}$ samples the next frame using original-image flow. RGB values lie in $[0, 1]$; squared errors sum three channels and average valid pixels at the dataset's training-loading resolution. Smoothing can also reduce the error.

D. Aggregation and Visual Sampling

For a per-image metric m, a scene score is

$$\overline{m}_s = \frac{1}{300}\sum_{v=1}^{3}\sum_{t=1}^{100} m_{s,t,v}. \tag{S13}$$

Temporal error instead averages the $3 \times 99 = 297$ adjacent pairs. Each pair remains within the same camera sequence; the end of one camera is never paired with the beginning of another. nuScenes weights its ten scenes equally, $\overline{m} = \frac{1}{10}\sum_s \overline{m}_s$. This aggregation first produces a scene-level result and then a dataset-level mean.

No Top-30 or post-hoc resolution selection is used. Temporal montages show fixed frames 49, 54, 59, 64, and 69, separate from full-sequence aggregation. These images illustrate temporal behavior but do not replace the complete set used for scoring.

E. Reference Images and Interpretation

All content comparisons start from the same original observation. For Qwen-Image, J^w is the edited image itself; for scene methods, it is the rendering at the corresponding camera and time. The teacher is therefore one evaluated output, not the content reference for IGSW alone. CLIP-S favors content similarity, CLIP-DS rewards the prescribed weather change, Sem-CS tracks semantic regions, and temporal error tracks motion-aligned stability. Their complementary roles motivate joint reporting: retaining an original-looking image or smoothing an edited sequence can improve an individual metric without producing a better weather scene.

F. Shared Motion Reference

Flow is estimated from original observations so that all methods are evaluated with the same motion estimate, rather than allowing each edited sequence to change its own alignment. Forward–backward checking removes inconsistent correspondences, and sampling coordinates outside the image are excluded. The remaining mask is reused for every method at the same scene, timestamp, and camera. Large teacher-generated structures can therefore increase error without changing which pixels count as valid for that method.

The warping operator compares corresponding image content after camera and object motion, not unaligned pixel differences. It can still penalize weather phenomena whose motion differs from the original scene. Our main evaluations consequently use the trained weather appearance before additional particle rendering. Since the metric averages the squared Euclidean RGB norm, summing rather than averaging the three color channels is part of the protocol and must be held fixed in comparisons.

G. Training and Evaluation Membership

The full evaluation includes all synchronized observations used by Stage I. For Stage II, every fifth timestamp is withheld from weather refinement across all three cameras. This provides 20 post-training diagnostic timestamps, but does not make them novel observations for the complete two-stage model. Main tables report the same full sequence for every method, not only the refinement subset. The distinction keeps scene reconstruction, post-training behavior, and temporal consistency separate in the interpretation of the results.

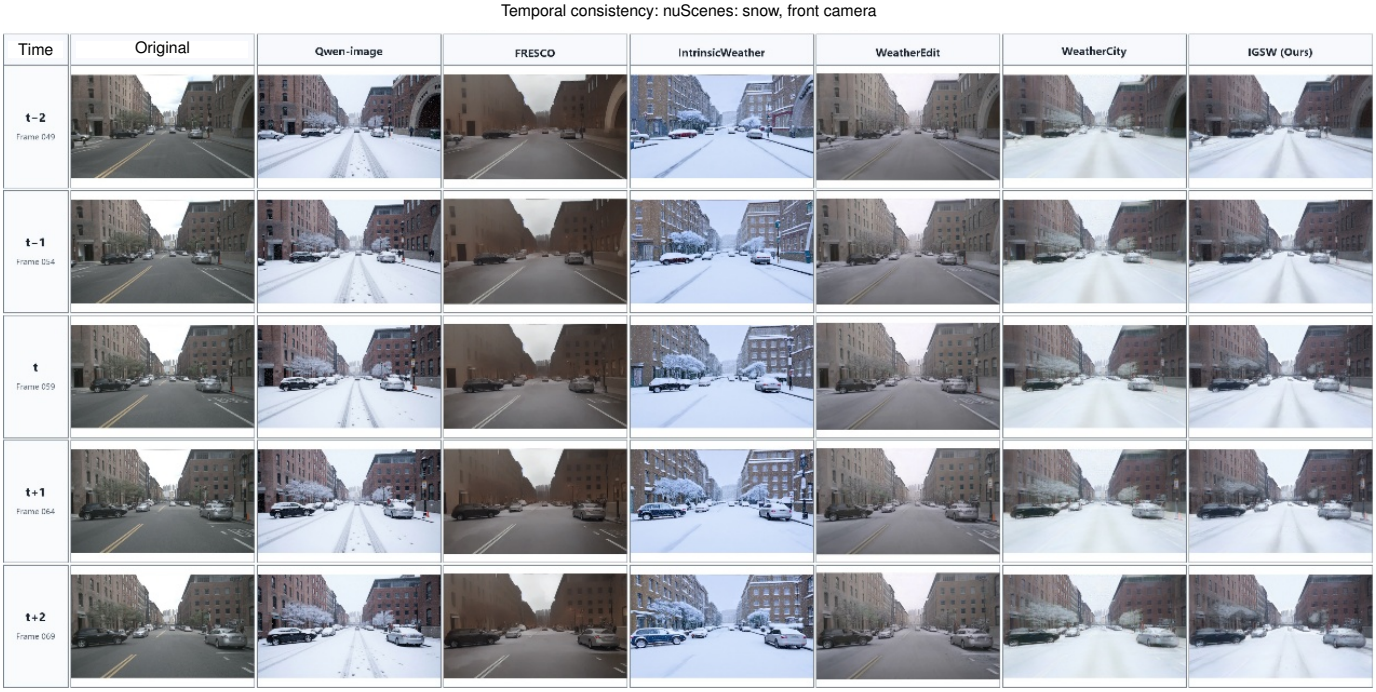


Fig. S1. Temporal snow comparison on nuScenes at frames 49, 54, 59, 64, and 69. The layout and method order match the main rain comparison. The montage illustrates fixed samples; full-sequence temporal evaluation uses all 297 adjacent pairs.

S5. Additional nuScenes Temporal Comparisons

A. Snow Appearance Across Time

Figure S1 complements the main-paper rain sequence with snow. Each column uses the same timestamp for all methods, and the original row supplies a content reference. The comparison is intended to show whether snow tone and surface appearance evolve coherently as the camera and traffic move. Frame selection follows the fixed sequence used for rain, not an ordering by per-image scores.

The Qwen teacher provides conspicuous target-weather appearance, but independently edited details can vary between neighboring observations. Scene optimization integrates these observations into a representation shared across time and views. The IGSW row should therefore be read together with the original and teacher rows: original content indicates what should remain, while the teacher indicates the intended weather change. Closer teacher resemblance is desirable only where the teacher has preserved valid scene content.

B. Reading Temporal and Structural Evidence

Temporal smoothness and structural fidelity are related but distinct. A stable overlay can have low frame-to-frame variation while still covering a building; conversely, strongly varying illumination can increase temporal error without changing semantic identity. The main paper consequently reports CLIP-S, CLIP-DS, Sem-CS, and motion-aligned error together, rather than treating a single metric as a complete measure of weather-editing quality.

The structure-preservation example in main Fig. 5 addresses the former case. Its cloud-covered building motivates the use of the original observation as a spatial reference in conflicted regions. Color-statistic transfer preserves a weather-compatible tone there, while the gated edge objective discourages the loss of local boundaries. Outside those regions, the safe teacher retains the original edited weather image, allowing stronger road, sky, and illumination changes.

C. Relation to the Ablations

The module ablation separates the initial appearance representation from subsequent refinement. Stage I alone changes metrics modestly, whereas Stage II produces more noticeable teacher-style transfer. The full model gives the highest three content/direction/semantic scores in that ablation, while Stage II alone has lower temporal error. Thus, the contribution of the first stage is not simply to minimize frame-to-frame differences.

The component ablation further separates content sharing, weather-specific decoding, the structure gate, and transport. In particular, removing the gate increases snow CLIP-DS but decreases semantic consistency, whereas removing transport reduces weather-direction alignment while lowering temporal error. These observed trade-offs explain why the complete method is evaluated for the combination of weather expression and retained content. The supplementary sequences add temporal context to the static comparisons; they do not introduce a different evaluation subset or a new quantitative experiment.

S6. Additional Temporal Comparisons

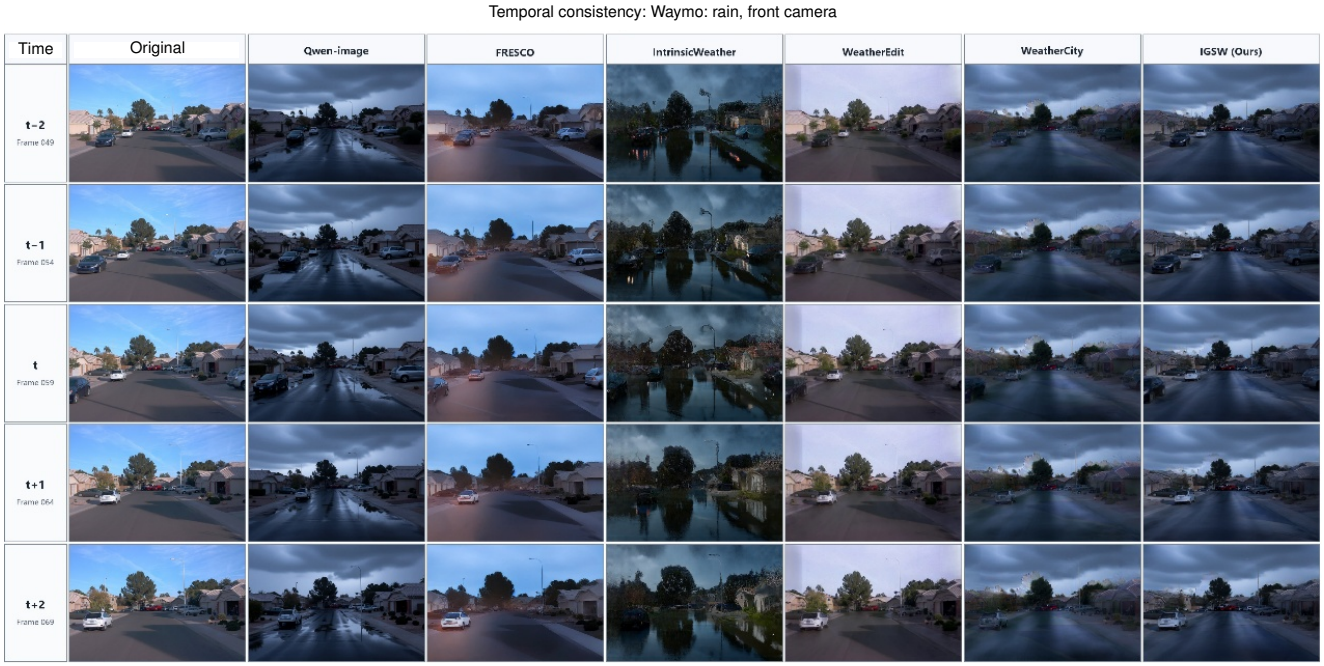


Fig. S2. Waymo rain temporal comparison. Columns show frames 49, 54, 59, 64, and 69. Method order matches the main paper.

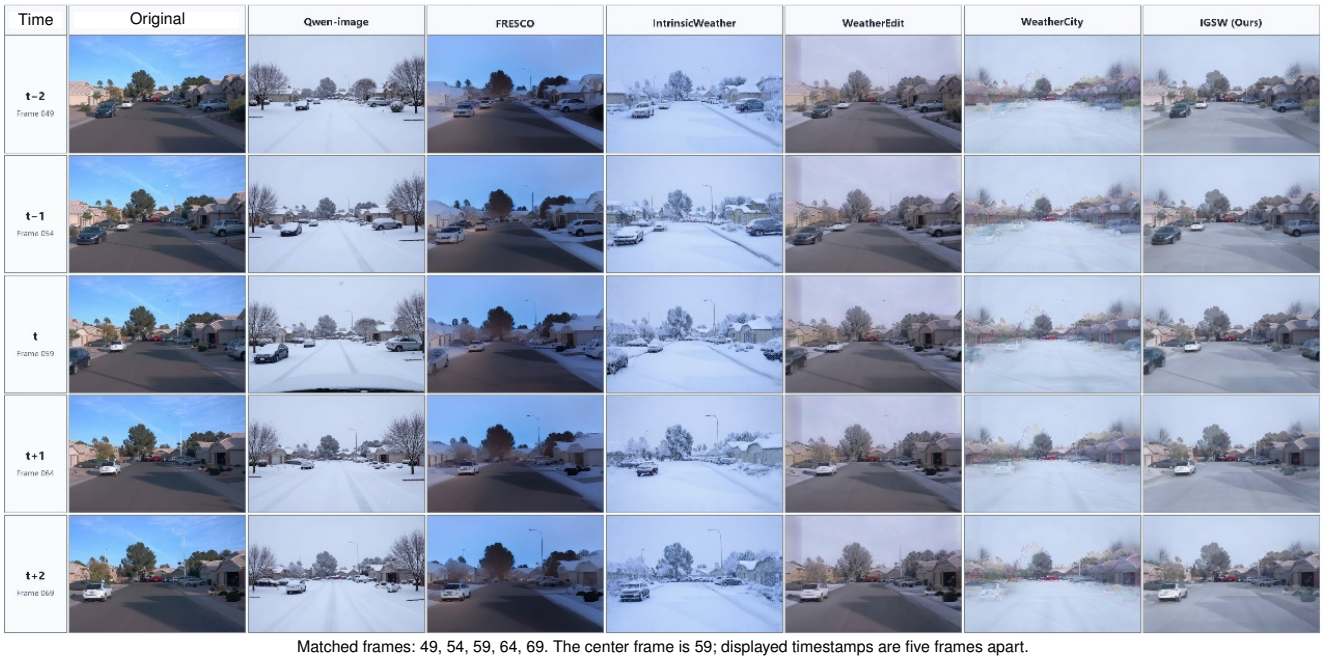


Fig. S3. Waymo snow temporal comparison at the same timestamps and with the same method order as the rain sequence.

Figures S2 and S3 retain the original Waymo temporal panels. Image teachers provide strong weather appearance but exhibit local frame-to-frame variation. IGSW transfers the target atmosphere into a shared scene representation. These fixed examples complement complete-sequence scores; they are not selected by per-frame metric rank.